

Prepare Your Laptop for the Labs

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Operating Systems, FCIM / FAF, UTM, 2026-2027. One script, run once, about 10 minutes. Do this **before** Lab 3.

What you need

- A laptop with Linux: Ubuntu, Debian, Mint, Pop!_OS, Fedora, Arch, Manjaro or openSUSE.
- Internet access and your password (the script asks for it once, to install packages).
- The file **os2026-setup.sh** from the course channel, saved in your Downloads folder.
- No Linux on your laptop? Tell the assistant **before** the lab: a Linux virtual machine or a live USB works too.

Three steps

1. Open a terminal (**Ctrl+Alt+T** on most systems) and go to the folder where you saved the file:

```
cd ~/Downloads      # or ~/Descarcari, wherever you saved it
```

2. Run the script and wait. It installs the RISC-V compiler and QEMU, downloads xv6 into ~/xv6-riscv, builds it and boots it once to test it:

```
bash os2026-setup.sh
```

3. The last line must be green: **All set**. From now on, start xv6 with:

```
cd ~/xv6-riscv && make qemu  
# to leave xv6: press Ctrl-A, release, then press X
```

On Ubuntu 22.04, Mint 21 and Debian 11 the script also builds a newer QEMU (the one these systems ship is too old for xv6). This takes 5 to 10 minutes more; let it run.

Safe to repeat

Run the script again whenever something looks wrong. It never deletes your work in ~/xv6-riscv. To only test, without installing anything:

```
bash os2026-setup.sh --check
```

If something goes wrong

You see	Do this
Unknown distribution	Show it to the assistant: the four tools (git, make, a riscv64 gcc, qemu-system-riscv64) are installed by hand.
sudo asks for a password, or “not in the sudoers file”	Type your login password (nothing is shown while you type). Without admin rights, ask the assistant.
“Could not get lock” (apt)	The system is updating in the background. Wait two minutes and run the script again.
“Failed to get write lock” (QEMU)	Another xv6 is still running. Close it (Ctrl-A, X) or run <code>pkill qemu-system-riscv64</code> , then retry.
git clone failed	Check the internet connection (try opening github.com), then run the script again.